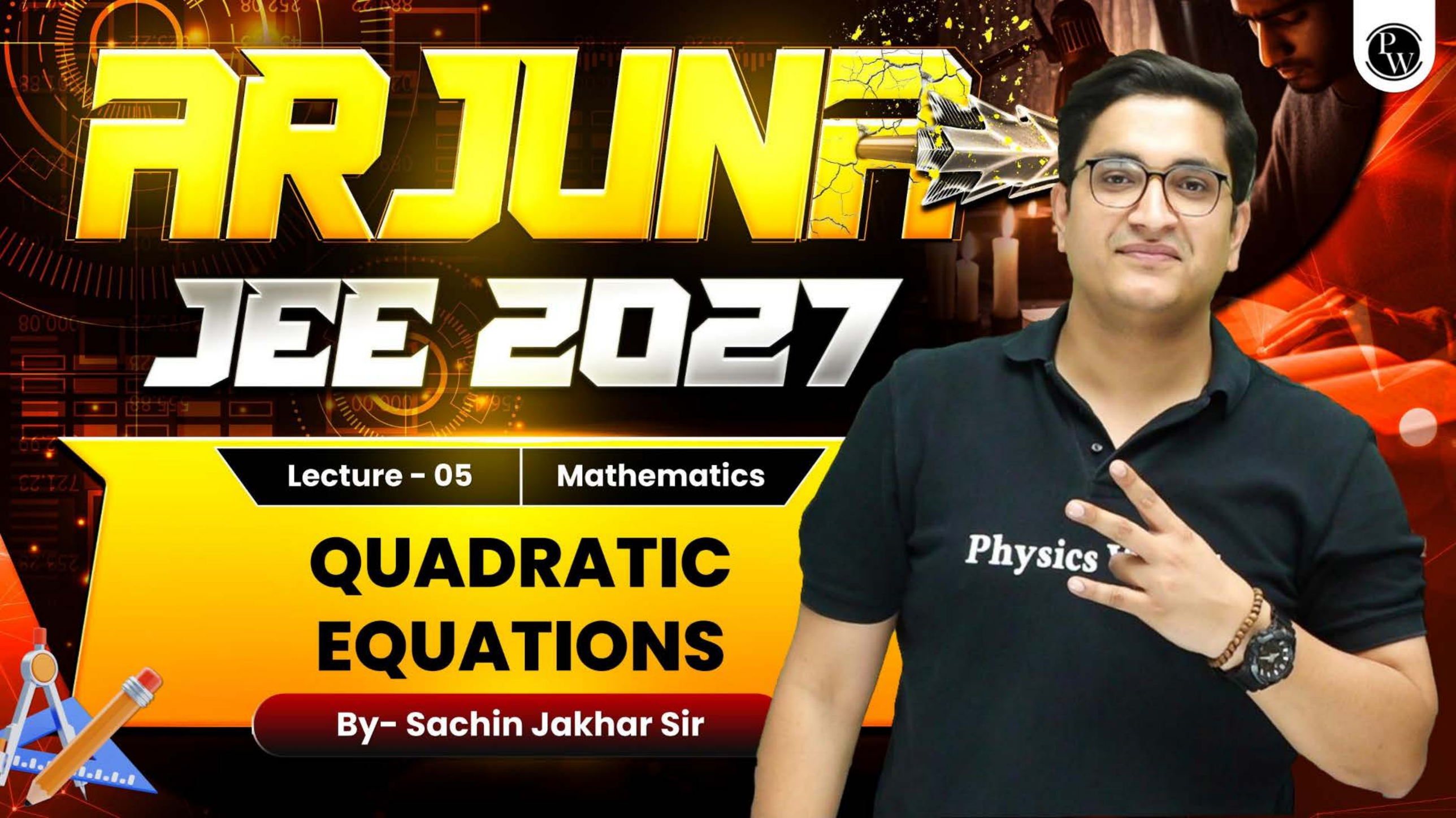
ARJUNA JEE 2027

Lecture - 05 | Mathematics

QUADRATIC EQUATIONS

By- Sachin Jakhar Sir



Topics *to be covered*

A Nature of Roots

B Types of Roots

C Question Practice

D





LAST CLASS REVISION

Newton's Formula:

Quadratic : $3x^2 - 5x - 32 = 0$ **with roots** α **&** β

$S_n = 3\alpha^n - 4\beta^n, n \geq 1$

Then:

$\Rightarrow \frac{3S_{n+2} - 5S_{n+1}}{8S_n} =$ 4 .

$$\frac{3S_{n+2} - 5S_{n+1}}{8S_n} = 4.$$

Question Practice

Question [JEE Mains-2021]

67%

If α, β are roots of the equation $x^2 + 5(\sqrt{2})x + 10 = 0$, $\alpha > \beta$ and $P_n = \alpha^n - \beta^n$ each positive integer n, then the value of $\left(\frac{P_{17}P_{20}+5\sqrt{2}P_{17}P_{19}}{P_{18}P_{19}+5\sqrt{2}P_{18}^2}\right)$ is equal to _____.

$x^2 + 5\sqrt{2}x + 10 = 0 \begin{cases} \alpha \\ \beta \end{cases}$

$\# \quad P_n + 5\sqrt{2}P_{n-1} + 10P_{n-2} = 0$

$n = 20 \Rightarrow P_{20} + 5\sqrt{2}P_{19} + 10P_{18} = 0$

$n = 19 \Rightarrow$

$$\frac{P_{17}(P_{20} + 5\sqrt{2}P_{19})}{P_{18}(P_{19} + 5\sqrt{2}P_{18})} = \frac{P_{17}(-10P_{18})}{P_{18}(-10P_{17})} = 1$$

Question [JEE Mains-2023]



49%

If a and b are the roots of equation $x^2 - 7x - 1 = 0$, then the value of $\frac{a^{21} + b^{21} + a^{17} + b^{17}}{a^{19} + b^{19}}$ is equal to

a

b

$S_n = a^n + b^n$

$S_n - 7S_{n-1} - S_{n-2} = 0$

$n=21$: $S_{21} - 7S_{20} - S_{19} = 0$

$n=19$: $S_{19} - 7S_{18} - S_{17} = 0$

$n=20$: $S_{20} - 7S_{19} - S_{18} = 0$

$$\frac{S_{21} + S_{17}}{S_{19}} = \frac{(7S_{20} + S_{19}) + S_{17}}{S_{19}}$$

$$= \frac{7(7S_{19} + S_{18}) + S_{19} + S_{17}}{S_{19}}$$

$$= \frac{49S_{19} + 7S_{18} + S_{19} + S_{17}}{S_{19}} = \frac{50S_{19} + S_{19} - S_{17} + S_{17}}{S_{19}} = \frac{51S_{19}}{S_{19}} = 51$$

M-2

$$N^r = a^{21} + b^{21} + a^{17} + b^{17}$$

$$= a^{17}(a^4+1) + b^{17}(b^4+1)$$

$$= a^{17}(51a^2) + b^{17}(51b^2)$$

$$\frac{N^r}{a^{19}+b^{19}} = \frac{51(a^{19}+b^{19})}{a^{19}+b^{19}} = 51$$

$x^2 - 7x - 1 = 0$: roots a, b

$a^2 - 7a - 1 = 0$

SBS

$a^2 - 1 = 7a$

$a^4 + 1 - 2a^2 = 49a^2$

$a^4 + 1 = 51a^2$

Question [JEE Mains-2023]

$\alpha^7+\beta^7=(\alpha^3+\beta^3)(\alpha^4+\beta^4)-(\alpha^4\beta^3+\beta^3\alpha^4)$

Let α, β be the roots of the equation $x^2-\sqrt{2}x+2=0$. Then $\alpha^{14}+\beta^{14}$ is equal to

A $-64\sqrt{2}$

B $-128\sqrt{2}$

C -64

D -128

$\left\{\begin{array}{l}\alpha'+\beta'=\sqrt{2}\\ \alpha\beta=2\end{array}\right\}$

SBS

$\alpha^2+\beta^2+2(\alpha\beta)=2$

$\alpha^2+\beta^2=-2$.

$\alpha^2+\beta^2=-2=-2^1$

$\alpha^4+\beta^4=-4=-2^2$

$\alpha^8+\beta^8=-16=-2^4$

Cube.

$\alpha^{14}+\beta^{14}=(\alpha^{16}+\beta^{16})(\alpha^{-2}+\beta^{-2})-(\beta^{16}\alpha^{-2}+\alpha^{16}\beta^{-2})$

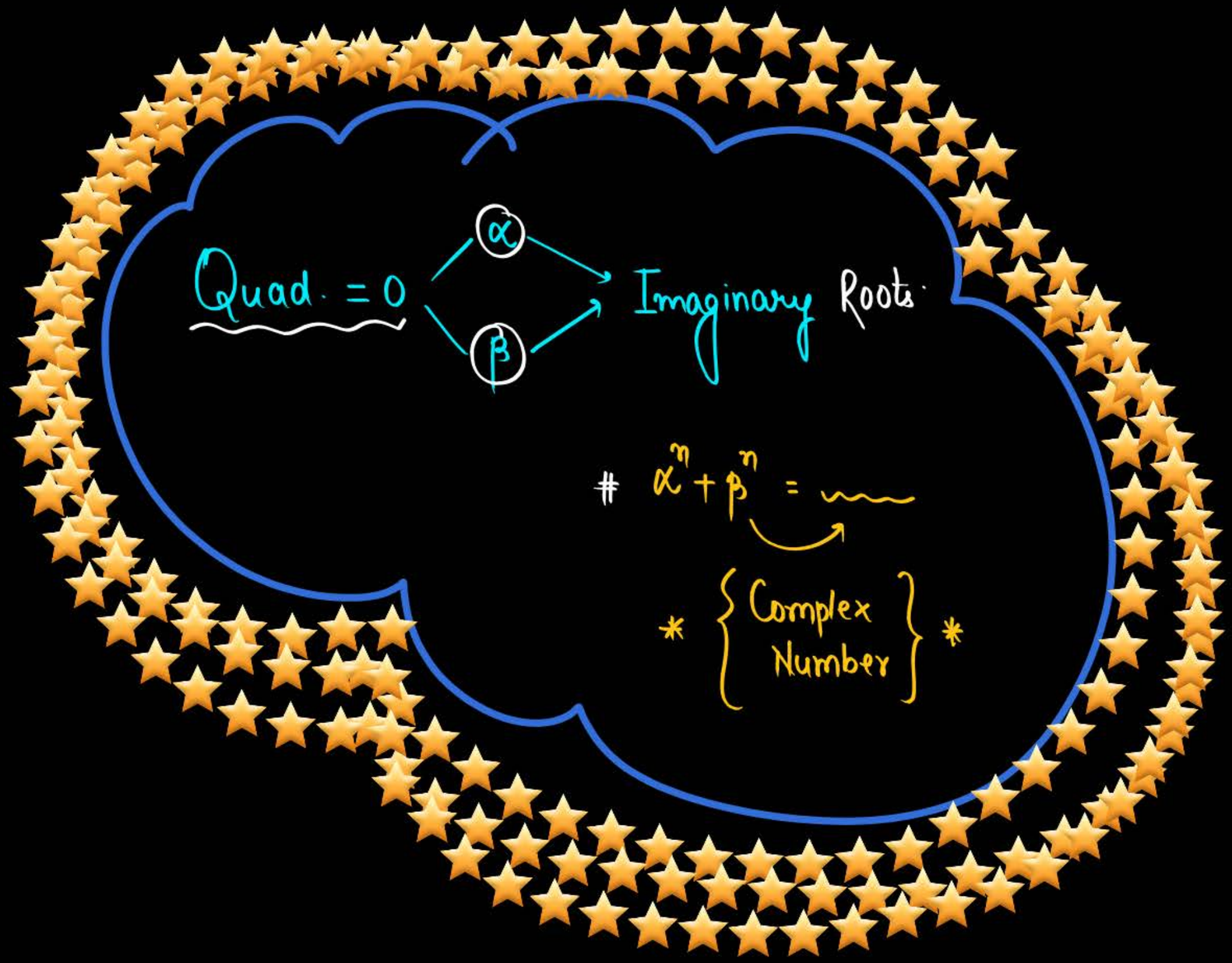
$\alpha^{14}+\beta^{14}=(\alpha^8+\beta^8)(\alpha^6+\beta^6)-(\alpha^6\beta^8+\beta^6\alpha^8)$

$x^2 - \sqrt{2}x + 2 = 0$, $S_n = \alpha^n + \beta^n$.

$S_n - \sqrt{2}\,S_{n-1} + 2\,S_{n-2} = 0$.

$S_{14} - \sqrt{2}\,S_{13} + 2\,S_{12} = 0$.

⋮


Quad. = 0
α
β
Imaginary Roots
$\alpha^n + \beta^n$ = ~~~
* { Complex Number } *

Question [JEE Mains-2023]

Let α, β be the roots of the quadratic equation $x^2 + \sqrt{6}x + 3 = 0$. Then

$\frac{\alpha^{23} + \beta^{23} + \alpha^{14} + \beta^{14}}{\alpha^{15} + \beta^{15} + \alpha^{10} + \beta^{10}}$ is equal to



A 729

B 72

C 81

D 9



NATURE OF ROOTS

CHAUDHARY kaun hai : $D = b^2 - 4ac$

Quadratic: $ax^2 + bx + c = 0, (a, b, c \in R), a \neq 0$

1. For Roots to be Real and distinct. $\Rightarrow D > 0$
2. For Roots to be Real and Equal. $\Rightarrow D = 0$
3. For Imaginary (or Non-Real) Roots. $\Rightarrow D < 0$
4. For Real Roots. $\Rightarrow D \geq 0$
5. For Quadratic to become perfect square. $\Rightarrow D = 0$

Example

Find all values of the parameter a for which the quadratic equation $(a+1)x^2 + 2(a+1)x + a - 2 = 0$ has

(a) two distinct roots,

(b) no real roots

(c) two equal roots.

coeff. of $x^2 \neq 0$

$a+1 \neq 0$

$a \neq -1$

$D = (2(a+1))^2 - 4(a+1)(a-2)$

$= 4(a+1)^2 - 4(a+1)(a-2)$

$= 4(a+1)((a+1) - (a-2))$

$D = 12(a+1)$

(a) $D > 0$

$12(a+1) > 0$

$a > -1$

(b) $D < 0$

$12(a+1) < 0$

$a < -1$

(c) $D = 0$

$12(a+1) = 0$

$a = -1$

n.p.

ϕ ✓

Question [JEE Mains-2020]

The least positive value of 'a' for which the question,

$2x^2 + (a - 10)x + \frac{33}{2} = 2a$ has real roots is

$D \geq 0$

.new.

(80% DIBY)

Module Question

Let $p, q \in \{1, 2, 3, 4\}$. The number of equations of the form $px^2 + qx + 1 = 0$ having real roots is

(A) 15

(B) 9

(C) 7

(D) 8

$D \geq 0$

$q^2 - 4p \geq 0$

$q^2 \geq 4p$

$p = 1 \Rightarrow q^2 \geq 4 \Rightarrow q = 2, 3, 4$

$p = 2 \Rightarrow q^2 \geq 8 \Rightarrow q = 3, 4$

$p = 3 \Rightarrow q^2 \geq 12 \Rightarrow q = 4$

$p = 4 \Rightarrow q^2 \geq 16 \Rightarrow q = 4$

$1x^2 + 2x + 1 = 0$

$1x^2 + 3x + 1 = 0$

$1x^2 + 4x + 1 = 0$

$2x^2 + 3x + 1 = 0$

$2x^2 + 4x + 1 = 0$

Total = 7

$p, q \in \{0, 1, 2, 3, 4\}$

$px^2 + qx + 1 = 0$

$D \geq 0$

$q^2 - 4p \geq 0$

$q^2 \geq 4p$

~~$p = 0$~~ $\Rightarrow$ $q^2 \geq 0$

$q = 0, 1, 2, 3, 4.$

$p = 1 \Rightarrow q^2 \geq 4 \Rightarrow q = 2, 3, 4$

$p = 2 \Rightarrow q^2 \geq 8 \Rightarrow q = 3, 4.$

$p = 3 \Rightarrow q^2 \geq 12 \Rightarrow q = 4$

$p = 4 \Rightarrow q^2 \geq 16 \Rightarrow q = 4$

Question

70%

Let λ_1 and λ_2 be two values of λ for which the expression $x^2 + (2-\lambda)x + \lambda - \frac{3}{4}$ becomes a perfect square. The value of $(\lambda_1^2 + \lambda_2^2)$ equals

$D = 0$

$(2-\lambda)^2 - 4\left(\lambda - \frac{3}{4}\right) = 0$.

$\lambda^2 + 4 - 4\lambda - 4\lambda + 3 = 0$

$\lambda^2 - 8\lambda + 7 = 0$ $\begin{cases} 7 = \lambda_1 \\ 1 = \lambda_2 \end{cases}$

$49 + 1 = 50$.

A 8

B 25

C 50

D 100

Question

For what value of p, $(4-p)x^2+(2p+4)x+(8p+1)=0$ is a perfect square.

DIBY - A

Ans.

Acidic Question

If p & q are real numbers such that quadratic $px^2 + qx + 2 = 0$ vanishes for $x = 1$ then roots of quadratic : $3px^2 + 5qx + 14 = 0$ are

A. Positive

B. Negative

C. Real & distinct

D. Imaginary

Root $x = 1$

vanishes → becomes zero

$x = 1$

$p + q + 2 = 0$

$\# \quad p = -q - 2$

$D = (5q)^2 - 4 \times (3p)(14)$

$D = 25q^2 - 168p$

$= 25q^2 - 168(-q - 2)$

$D = 25q^2 + 168q + 2 \times 168$

+ve always

$\# \quad D > 0$

WCM

$25x^2 + 168x + 2 \times 168 > 0$



always +ve.

Challenger Question

Let $a, b \in R, a \neq 0$ be such that the equation, $ax^2 - 2bx + 5 = 0$ has a repeated root α, which is also a root of the equation, $x^2 - 2bx - 10 = 0$. If β is the other root of this equation, then $\alpha^2 + \beta^2$ is equal to:

"both roots are α"

diw.

A 25

B 28

C 26

D 24

Sachin Sir Special Summary (S^4):

$ax^2 + bx + c = 0$ $(a, b, c \in R)$, $(a \neq 0)$

- $D = 0$ (Roots are real & equal)
- $D > 0$ (Roots are real & distinct)
  - $a, b, c \in Q$ ✓
    $D \equiv$ Perfect square ✓
    (Roots are Rational) ✓
    - $a = 1, b$ & $c \in I$
      $D \equiv$ Perfect square
      (Roots are Integral)
  - $a, b, c \in Q$
    $D \equiv$ Not a perfect square
    (Roots are Irrational) ✓
    $p + \sqrt{q}, p - \sqrt{q}$
    Pair में आएंगे
- $D < 0$ (Roots imaginary)
  $(p + iq, p - iq)$
  # $3 + 2i$, $3 - 2i$.
  Pair में exist करते ही है!

DO IT BY YOURSELF (DIBY)

DIBY-07

If α, β are the roots of $ax^2 + bx + c = 0, (a \neq 0)$ and $\alpha + \delta, \beta + \delta$ are the roots of $Ax^2 + Bx + C = 0, (A \neq 0)$ for some constant δ, then prove that, $\frac{b^2-4ac}{a^2} = \frac{B^2-4AC}{A^2}$.

[JEE-2000]

DIBY-08

If α and β be two roots of the equation $x^2 - 64x + 256 = 0$. Then the value of $\left(\frac{\alpha^3}{\beta^5}\right)^{\frac{1}{8}} + \left(\frac{\beta^3}{\alpha^5}\right)^{\frac{1}{8}}$ is

(A) 1 (B) 3 (C) 4 (D) 2

[Ans. : (D)]

DIBY-09

Let α, β be the roots of the equation $x^2 + 2\sqrt{2}x - 1 = 0$. The quadratic equation, whose roots are $\alpha^4 + \beta^4$ and $\frac{1}{10}(\alpha^6 + \beta^6)$, is: [JEE Mains-2024]

(A) $x^2 - 190x + 9466 = 0$ (B) $x^2 - 195x + 9466 = 0$

(C) $x^2 - 195x + 9506 = 0$ (D) $x^2 - 180x + 9506 = 0$

[Ans. : (C)]

DIBY-10

If the equation $a(b-c)x^2+b(c-a)x+c(a-b)=0$ has equal roots, where $a+c=15$ and $b=\frac{36}{5}$, then a^2+c^2 is equal to [JEE Mains-2025]

[Ans. : 117]

DIBY-11

Let $\alpha \in \mathbb{R}$ and let α, β be the roots of the equation $x^2+60\sqrt[4]{x}+a=0$.
If $\alpha^4+\beta^4=-30$, then the product of all possible values of a is [JEE Mains-2023]

[Ans. : 45]

DIBY-12

If 2 and 6 are the roots of the equation $ax^2+bx+1=0$, then the quadratic equation, whose roots are $\frac{1}{2a+b}$ and $\frac{1}{6a+b}$, is : [JEE Mains-2024]

(A) $x^2+8x+12=0$
(B) $2x^2+11x+12=0$
(C) $4x^2+14x+12=0$
(D) $x^2+10x+16=0$

[Ans. : (A)]

DIBY-13

Let α and β be the roots of the equation, $5x^2 + 6x - 2 = 0$. If $S_n = \alpha^n + \beta^n, n = 1, 2, 3 \ldots$., then [JEE Mains-2020]

(A) $6S_6 + 5S_5 = 2S_4$
(B) $6S_6 + 5S_5 + 2S_4 = 0$
(C) $5S_6 + 6S_5 = 2S_4$
(D) $6S_6 + 6S_5 + 2S_4 = 0$

[Ans. : (C)]

DIBY-14

If α, β are the roots of the equation, $x^2 - x - 1 = 0$ and $S_n = 2023\alpha^n + 2024\beta^n$, then [JEE Mains-2024]

(A) $2S_{12} = S_{11} + S_{10}$
(B) $S_{12} = S_{11} + S_{10}$
(C) $2S_{11} = S_{12} + S_{10}$
(D) $S_{11} = S_{10} + S_{12}$

[Ans. : (B)]

DIBY-15

Let α, β be roots of $x^2 + \sqrt{2}x - 8 = 0$. If $U_n = \alpha^n + \beta^n$, then $\frac{U_{10} + \sqrt{2}U_9}{2U_8}$ is equal to [JEE Mains-2024]

[Ans. : 4]

DIBY-16

Let $\alpha, \beta; \alpha > \beta$, be the roots of the equation $x^2 - \sqrt{2}x - \sqrt{3} = 0$. Let $P_n = \alpha^n - \beta^n, n \in N$. Then $(11\sqrt{3} - 10\sqrt{2})P_{10} + (11\sqrt{2} + 10)P_{11} - 11P_{12}$ is equal to:

[JEE Mains-2024]

(A) $10\sqrt{2}P_9$ (B) $10\sqrt{3}P_9$ (C) $11\sqrt{2}P_9$ (D) $11\sqrt{3}P_9$

[Ans. : (B)]

DIBY-17

Let α and β be the roots of $x^2 - 6x - 2 = 0$. If $a_e = \alpha^n - \beta^n$ for $n \geq 1$, then the value of $\frac{a_{10} - 2a_8}{3a_9}$ is:

[JEE Mains-2021]

(A) 2 (B) 1 (c) 4 (d) 3

[Ans. : (A)]

DIBY-18

Let $\alpha, \beta(\alpha > \beta)$ be the roots of the quadratic equation $x^2 - x - 4 = 0$. If $P_n = \alpha^n - \beta^n, n \in N$, then $\frac{P_{15}P_{16} - P_{14}P_{16} - P_{15}^2 + P_{14}P_{15}}{P_{13}P_{14}}$ is equal to.

[JEE Mains-2022]

[Ans. : 16]

QUADRATIC INVOLVING MODULUS (already done B.M.)

DIBY-19

The number of real solutions of the equation, $x^2 - |x| - 12 = 0$ is:

[JEE Mains-2021]

(A) 1 (B) 4 (C) 3 (D) 2

[Ans. : (D)]

DIBY-20

The sum, of the squares of all the roots of the equation $x^2 + |2x - 3| - 4 = 0$, is

[JEE Mains-2025]

(A) $6(2 - \sqrt{2})$ (B) $3(3 - \sqrt{2})$ (C) $6(3 - \sqrt{2})$ (D) $3(2 - \sqrt{2})$

[Ans. : (A)]

DIBY-21

The product of the roots of the equation $9x^2 - 18|x| + 5 = 0$, is:

[JEE Mains-2020]

(A) $\frac{25}{9}$ (B) $\frac{25}{81}$ (C) $\frac{5}{9}$ (d) $\frac{5}{27}$

[Ans. : (B)]

QUADRATIC INVOLVING MODULUS

DIBY-22

The sum of the squares of the roots of $|x+2|^2+|x-2|-2=0$ and the squares of the roots of $x^2-2|x-3|-5=0$, is [JEE Mains-2025]

(A) 26 (B) 36 (C) 30 (d) 24

[Ans. : (B)]

DIBY-23

The number of real roots of the equation $x|x|-5|x+2|+6=0$

[JEE Mains-2023]

(A) 5 (B) 3 (C) 6 (D) 4

[Ans. : (B)]

DIBY-24

The sum of all the roots of the equation $|x^2-8x+15|-2x+7=0$ is:

[JEE Mains-2023]

(A) $9+\sqrt{3}$ (B) $11+\sqrt{3}$ (C) $9-\sqrt{3}$ (D) $11-\sqrt{3}$

[Ans. : (A)]

DIBY-25

The number of real solutions of the equation $x|x+5|+2|x+7|-2=0$ is

[JEE Mains-2024]

[Ans. : 3]

DIBY-26

If p and q are non-zero real numbers and $\alpha^3+\beta^3=-p, \alpha\beta=q$, then a quadratic equation whose roots are $\frac{\alpha^2}{\beta}, \frac{\beta^2}{\alpha}$ is

(A) $qx^2+px+q^2=0$

(B) $px^2+qx+p^2=0$

(C) $qx^2-px+q^2=0$

(D) $px^2-qx+p^2=0$

[Ans. : (A)]

DIBY-27(Challenger)

Let $f(x)=ax^2+bx+c, a, b, c \in I$ & $f(1)=0, 50<f(7)<60$ & $70<f(8)<80$, then $f(-1)$ is equal to

[Ans. : 14]

DIBY-28

If roots of the equation $(x-\alpha)(x-\beta)+(x-\gamma)(x-\delta)=0$ are m & n respectively, then roots of the equation $2(x-m)(x-n)-(x-\alpha)(x-\beta)$ Will be

(A) $\frac{\alpha+m+\gamma}{2}, \frac{\beta+n+\delta}{2}$

(B) γ, δ

(C) $\alpha - m, \beta - n$

(D) $m-\alpha+\gamma, n-\beta+\delta$

[Ans. : (B)]



Homework

Re-attempt all Class Examples

DPP & DIBY

Module: Basic Maths (4 Sittings Each of 1 Hr.)

Exercise-03(BRIDGE): COMPLETE

Exercise-04(JEE ADV. LEVEL):COMPLETE

It is never about end result

It's always about Journey

Thank You

#futureIITians